

Masked Reconstruction Lab. Instructions

- Load the MNIST dataset
- Scale pixel values to 0 to 1
- Reshape images to 28 x 28 x 1
- Create a masked version of each image by covering the center with zeros
- Use the masked image as input
- Use the original clean image as target
- Build a simple convolutional autoencoder
- Compile the model with:
 - optimizer: Adam
 - loss: MSE
- Train for about 5 epochs
- Run predictions on the masked test images
- Plot:
 - original image
 - masked image
 - reconstructed image
- Compare results across several digits
- Answer:
 - Which digits were easiest to reconstruct?
 - Which digits were hardest?
 - Where did the model fail?
 - Why is this closer to inpainting than denoising?